Analytical Chemistry 1 (125101)

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Framework of this course

❖ Introduction (L1)

❖ Section I Chemical Measurements

1.1 Basic knowledge (L2)

1.2 Error and estimation (L3&L4)

1.3 Linear regression (L5)

❖ Section II Chemical Equilibrium

2.1 pH and buffer solutions (L6)

2.2 Titrations (L7&L8)

2.3 Potentiometry in titrations (L9)

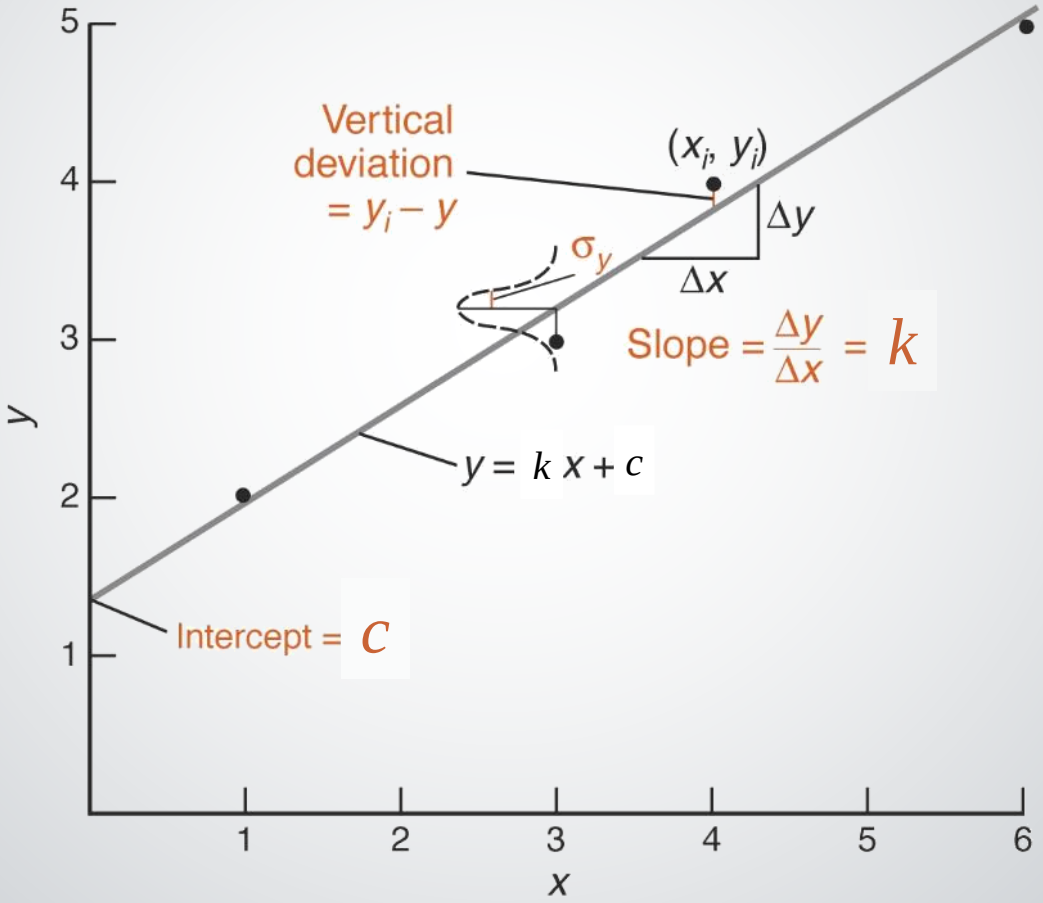
2.4 Gran plot in titrations (L10)

2.5 Electroanalysis (L11)

2.6 gravimetric analysis (L12)

❖ Summary (L13)

1.3 Linear regression



1.3 Linear regression analysis

Overview

1.3-1 Least squares method

1.3-2 Calibration methods

① External standard

② Standard addition

③ Internal standard

Linear regression analysis

We measure couples of interconnected variables $\{x_i, y_i\}$

x_i	y_i
6	8
7	18
8	32
9	37
10	54
11	58
12	66

Example:

x_i – Concentration

y_i – Signal reading

$$y = k\,x + c$$

Target: Find k and c

- Calculate y at any given x
- k and c have physical meaning

Linear regression analysis

x_i	y_i
6	8
7	18
8	32
9	37
10	54
11	58
12	66

Option A:

Solve 2 equations with 2 unknowns (k and c).

$$y_1 = k\,x_1 + c$$

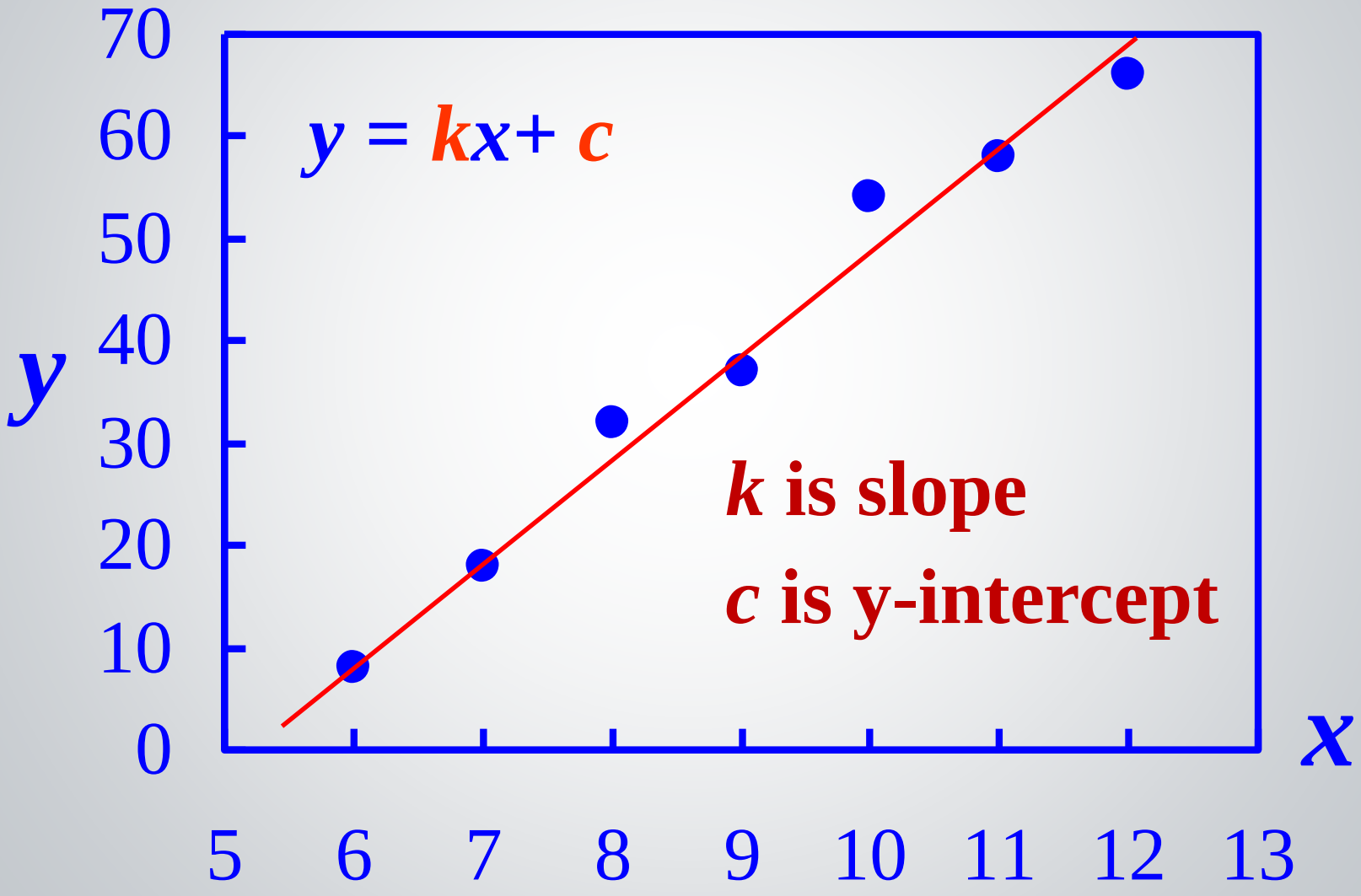
$$y_2 = k\,x_2 + c$$

$$8 = 6k + c$$

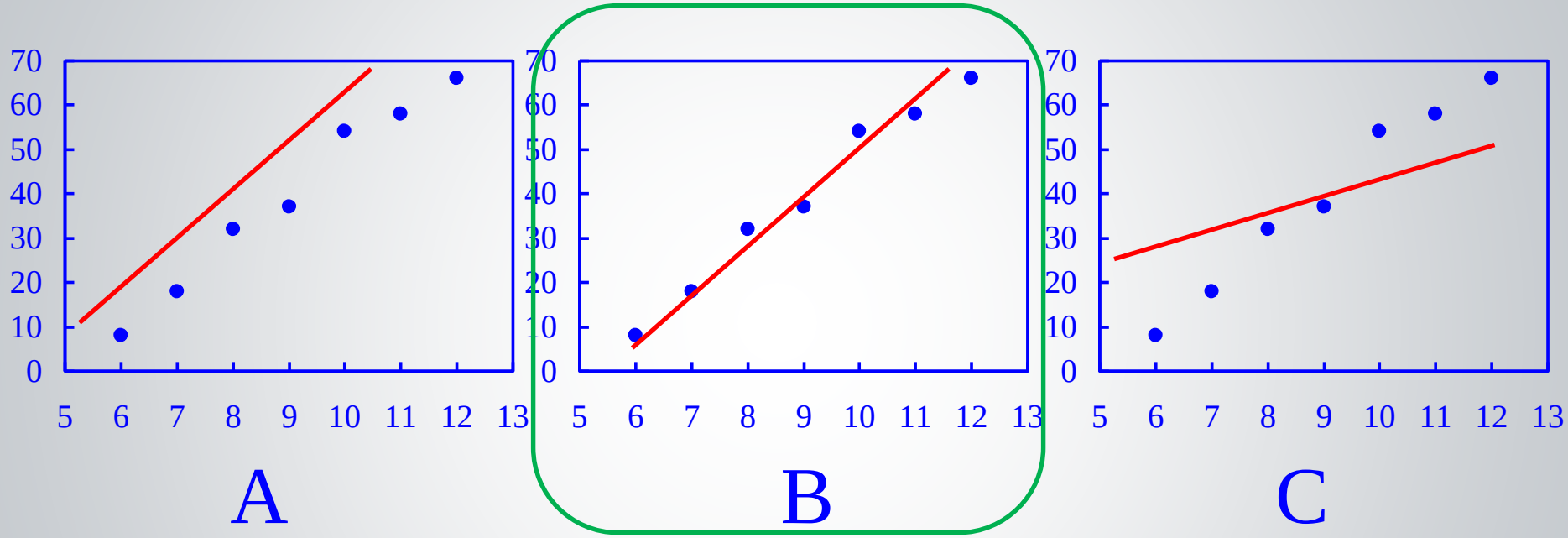
$$18 = 7k + c$$

$$y = 10\,x - 58$$

Option B: Linear regression



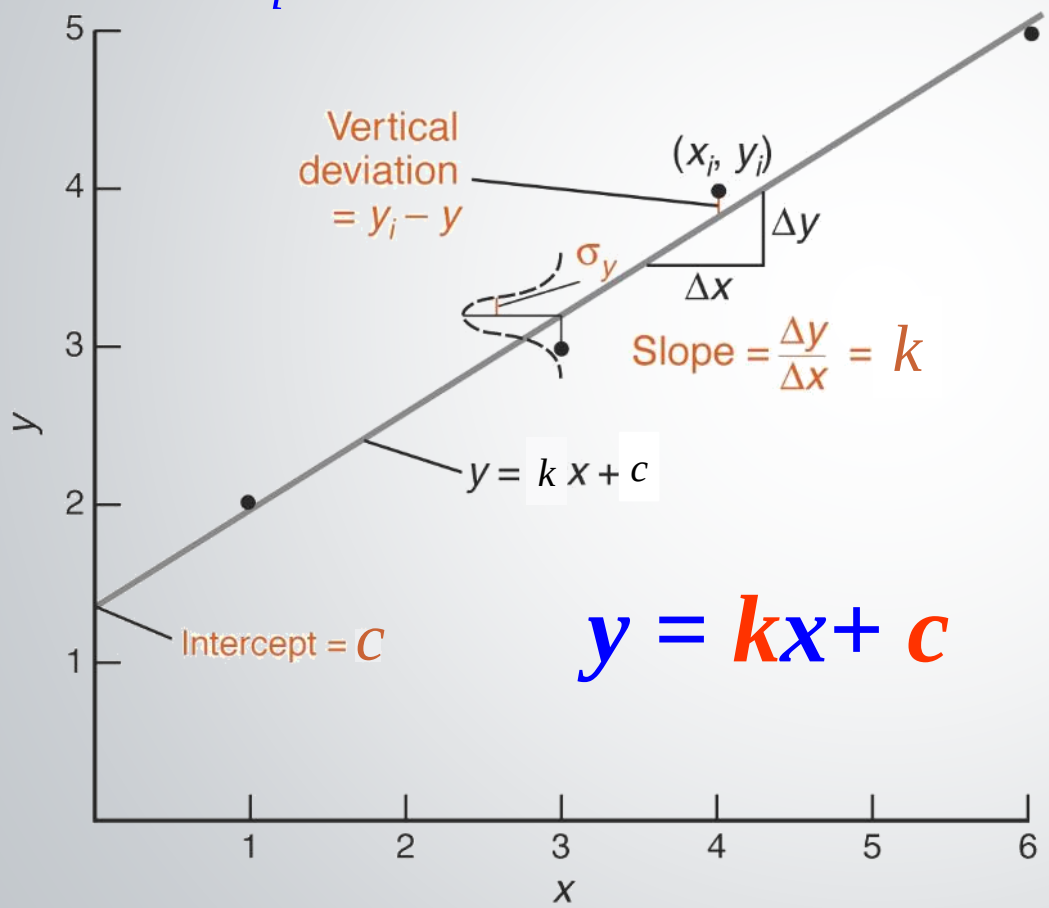
What is “the best straight line” fitting the experimental data points?



The line proving the maximum probability of getting the actual experimental results.

Probability of result y_i, at given k and c

$$\xi = \sum_i (y_i - c - kx_i)^2 \downarrow$$



$$y = kx + c$$

Assumption 1:
y is normally distributed

Assumption 2:
No errors in x

Assumption 3:
Each y_i result is independent of the others

<u>How to find the equation of the line?</u>

1.3-1 Method of least squares

The slope and intercept of the "best" straight line are found by calculus, (in form of determinant):

Least-squares "best" line

$$\text{Slope:} \quad k = \begin{vmatrix} \sum(x_i y_i) & \sum x_i \\ \sum y_i & n \end{vmatrix} \div D$$

$$\text{Intercept:} \quad c = \begin{vmatrix} \sum(x_i^2) & \sum(x_i y_i) \\ \sum x_i & \sum y_i \end{vmatrix} \div D$$

$$\text{where } D = \begin{vmatrix} \sum(x_i^2) & \sum x_i \\ \sum x_i & n \end{vmatrix}$$

$$k = \frac{N\left(\sum x_i y_i\right) - \left(\sum x_i\right)\left(\sum y_i\right)}{N\left(\sum x_i^2\right) - \left(\sum x_i\right)^2}$$

$$c = \frac{\left(\sum x_i^2\right)\left(\sum y_i\right) - \left(\sum x_i\right)\left(\sum x_i y_i\right)}{N\left(\sum x_i^2\right) - \left(\sum x_i\right)^2}$$

Least squares

$$\xi = \sum_i (y_i - c - kx_i)^2 \downarrow \qquad R^2 = \frac{[\sum(x_i - \bar{x})(y_i - \bar{y})]^2}{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2} \uparrow$$

x_i	y_i
6	8
7	18
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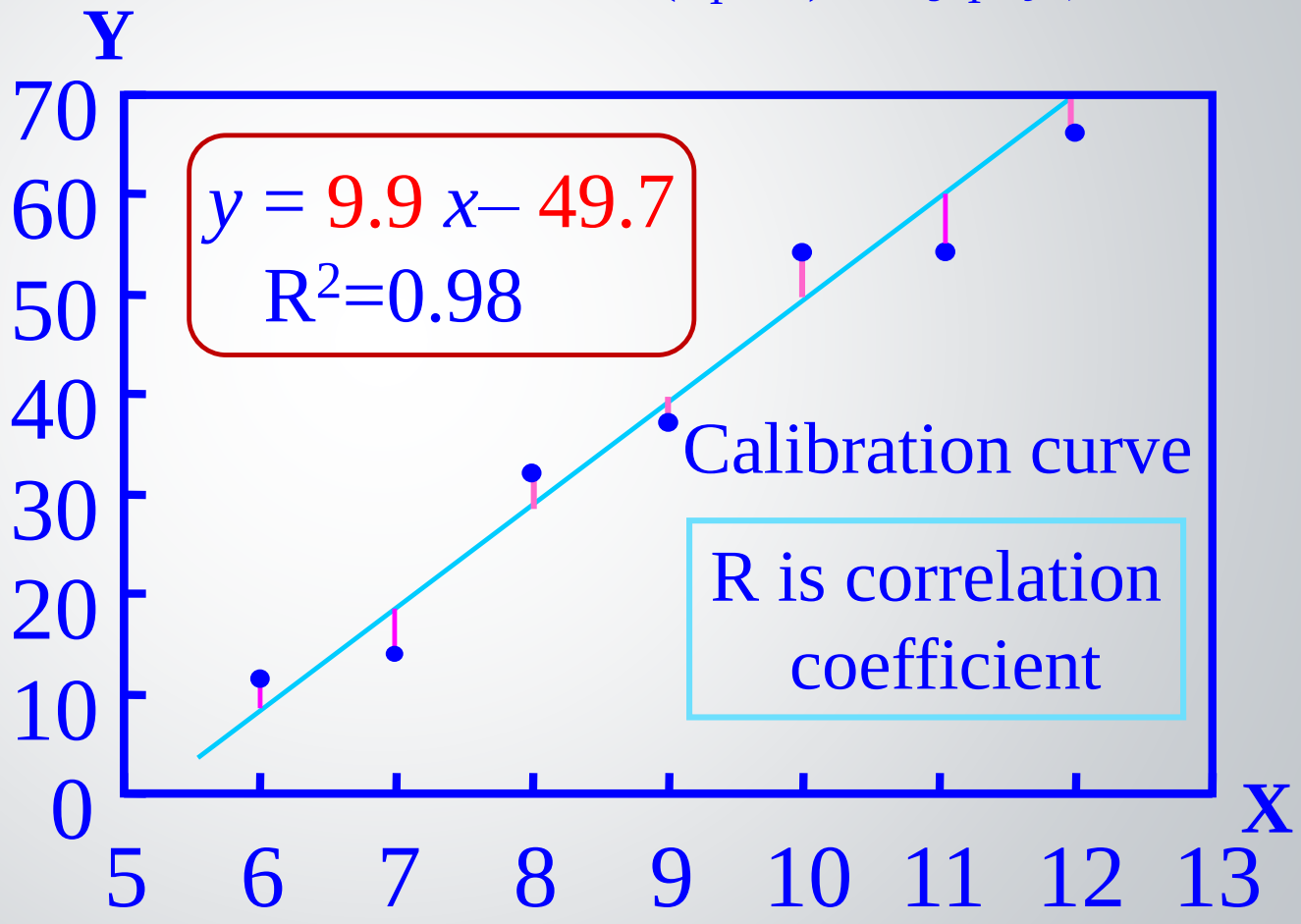


Option A: y = 10 x -58

1.3-2 Calibration methods

It is a method to measure the unknown quantity from the known quantities of analyte (standards)

① External standard

② Standard addition

③ Internal standard

① External standard

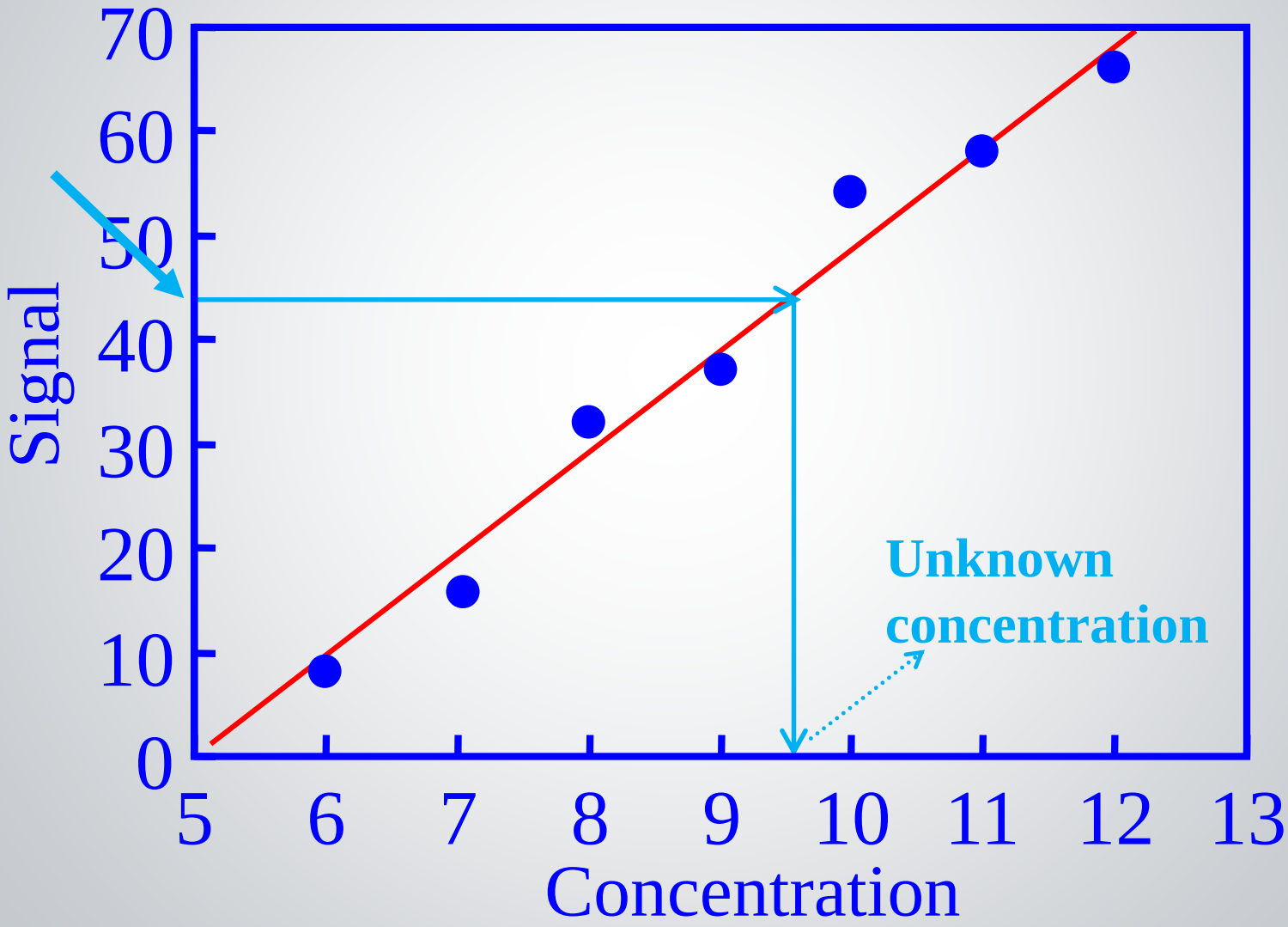
1) Prepare multiple solutions containing different [C]s of analyte with known quantities;

2) Measure signals for each [C];

3) Make the calibration curve;

4) Find the unknown [C] through the calibration curve.

① External standard



Matrix effect

Matrix: is everything in the sample, except analyte

Matrix effect: is the change in the analytical signal caused by anything in the sample other than analyte.

For example

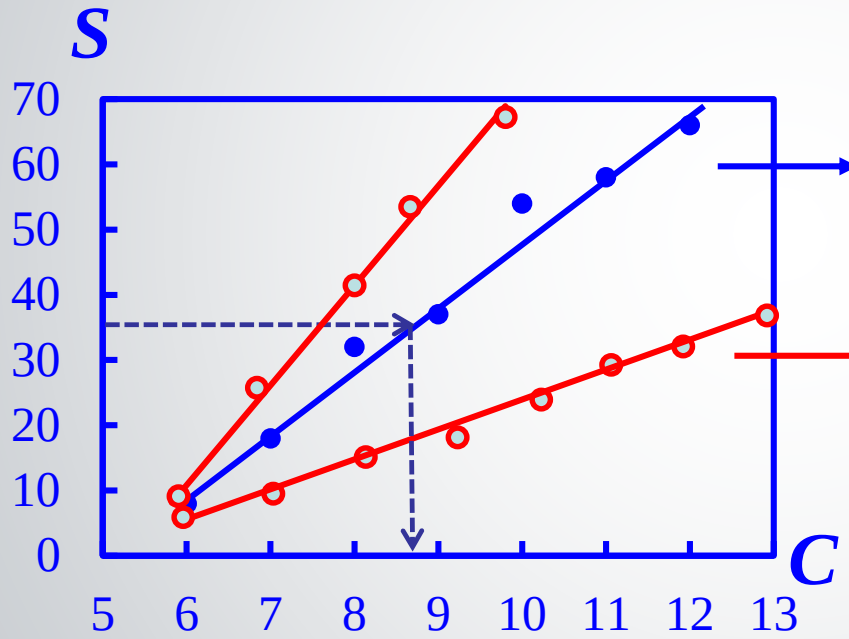
H3-4. The iodide (I^-) content of a commercial mineral water was measured by two methods. Method A found 0.23 milligrams of I^- per liter (mg/L) and method B found 0.0009 mg/L. When Mn^{2+} was added to the water of sample, the I^- concentration found by method A increased each time that more Mn^{2+} was added, but results from B were unchanged.

Which result is more reliable? What is occurring in these measurements and how to improve the analytical quality for the unreliable one?

Matrix effect

Matrix: is everything in the sample, except analyte

Matrix effect: is the change in the analytical signal caused by anything in the sample other than analyte.



Calibration curve of **pure** analyte

Calibration curve of analyte **in matrix**

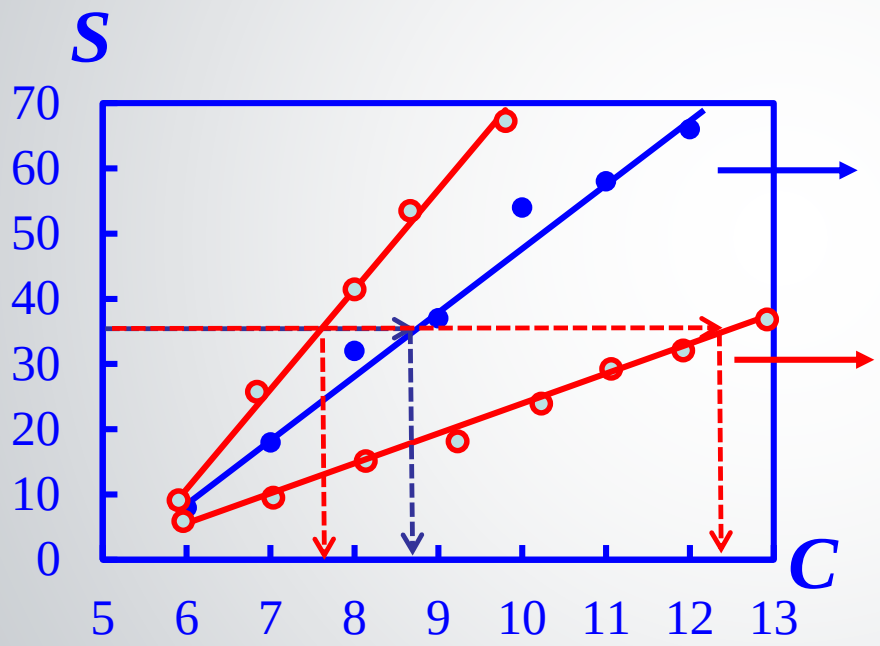
Slope becomes less or larger

- Matrix affects the magnitude of analytical signal;
- Calibration curve of pure analyte does not work!

Matrix effect

Matrix: is everything in the sample, except analyte

Matrix effect: is the change in the analytical signal caused by anything in the sample other than analyte.



Calibration curve of **pure** analyte

Calibration curve of analyte **in matrix**

Slope becomes less or larger

- Calibration curve of pure analyte does not work!

1.3-2 Calibration methods

It is a method to measure the unknown quantity from the known quantities of analyte (standards)

① External standard

② Standard addition

③ Internal standard

② Standard addition method

It is method base on **addition known** quantities of analyte to the unknown. From the increase of signal, deduce how much analyte is in the original.

- **Linear response**
- **Sample matrix is unknown or complex**

  It is impossible to create standards or blanks whose composition matches the unknown sample

- **Samples are made in almost the same matrix**

② Standard addition method

$$I_X = k\,[X]_i$$

$$I_{(X+S)} = k\,([X]_f + [S]_f)$$

$[X]_i$: initial concentration of unknown
I_X: X_i gives a signal intensity
$I_{(X+S)}$ is signal intensity after standard addition
$[X]_f$: final [C] of unknown in the solution
$[S]_f$: final [C] of standard in the solution

Small volume of concentrated standard addition don't change the matrix too much, so matrix composition is thought to be approximate, which means *k* is constant

$$\frac{I_X}{I_{(X+S)}} = \frac{k[X]_i}{k([X]_f+[S]_f)} = \frac{[X]_i}{[X]_f+[S]_f}$$

② Standard addition method

$$\frac{I_X}{I_{(X+S)}} = \frac{[X]_i}{[X]_f + [S]_f}$$

$$[X]_f = [X]_i \frac{V_o}{V}$$

$$[S]_f = [S]_i \frac{V_s}{V}$$

$[X]_i$: initial [C] of unknown

$[X]_f$: final [C] of unknown

$[S]_i$: initial [C] of standard

$[S]_f$: final [C] of standard

V_o is initial volume of unknown

V_s is added volume of standard

V is the total volume $V = V_o + V_s$

② Standard addition method

- ❑ One point of standard addition
- ❑ A solution with successive additions
- ❑ Multiple solutions with same final volume

❑ One point of standard addition

- Na^+ in serum gave a signal of 4.27 mV
- 5.00 mL of 2.08 M NaCl added to 95.0 mL serum sample
- The spiked serum gave a signal of 7.98 mV

Question: find the concentration of Na^+ in serum

$$[X]_f = [X]_i \frac{V_o}{V} = [X]_i \frac{95.0\ mL}{100.0\ mL} = 0.950\ [Na^+]_i$$

$\frac{V_o}{V}$ dilution factor

$$[S]_f = [S]_i \frac{V_s}{V} = 2.08\ M \frac{5.00\ mL}{100.0\ mL} = 0.104\ M$$

$$\frac{[X]_i}{[X]_f + [S]_f} = \frac{I_X}{I_{(X+S)}} \Rightarrow \frac{[Na]_i}{0.950[Na]_i + 0.104\ M} = \frac{4.27\ mV}{7.98\ mV}$$

$$[Na^+]_i = 0.113\ M$$

❑ A solution with successive additions

V_s, V and $I_{(X+S)}$ are variable

Other parameters are constant

$$\frac{[X]_i}{[X]_f+[S]_f}=\frac{I_X}{I_{(X+S)}}$$

$$[X]_f=[X]_i\frac{V_o}{V} \quad [S]_f=[S]_i\frac{V_s}{V}$$

$$\underbrace{\boxed{I_{(X+S)}\left(\frac{V}{V_o}\right)}}_{y}=I_X+\frac{I_X}{[X]_i}\underbrace{\boxed{[S]_i\left(\frac{V_s}{V_o}\right)}}_{x}$$

$$I_{(X+S)}=\frac{I_X}{[X]_i}\left([X]_f+[S]_f\right)$$

$$I_{(X+S)}=\frac{I_X}{[X]_i}\left([X]_i\frac{V_o}{V}+[S]_i\frac{V_s}{V}\right)$$

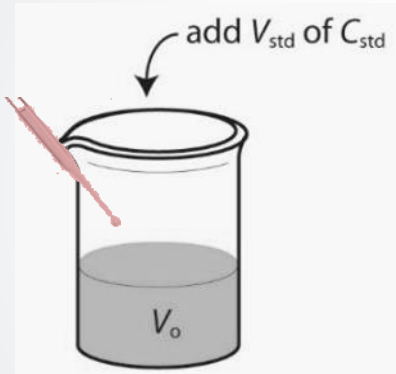
$$I_{(X+S)}=\left(I_X\frac{V_o}{V}+\frac{I_X}{[X]_i}[S]_i\frac{V_s}{V}\right)$$

$$y=I_X+\frac{I_X}{[X]_i}x$$

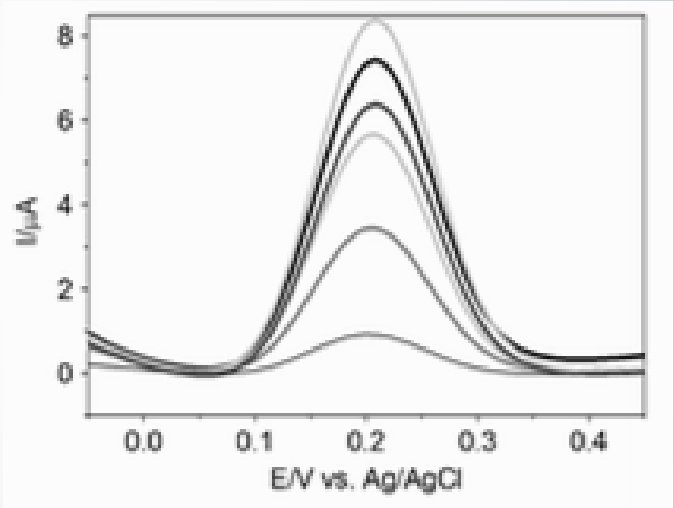
❑ A solution with successive additions

$$\underbrace{I_{(X+S)}\left(\frac{V}{V_o}\right)}_{y}=I_X+\frac{I_X}{[X]_i}\underbrace{[S]_i\left(\frac{V_s}{V_o}\right)}_{x}$$

Concentrated analyte



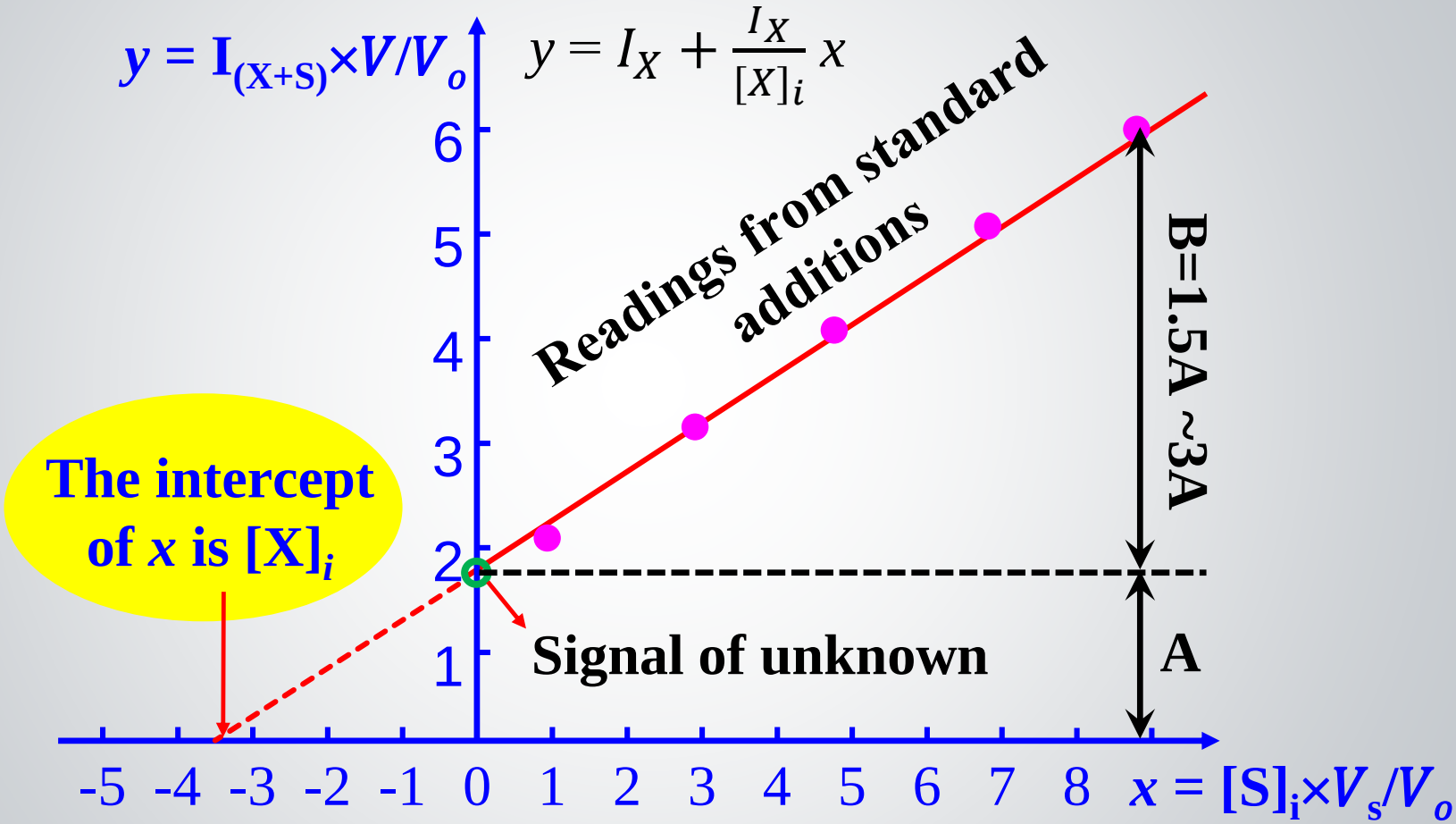
Unknown solution



V_o is constant, V_s and total volume V are variable, as a result, signal of $I_{(X+S)}$ changes,

$$\boldsymbol{I_{(X+S)}\left(\frac{V}{V_o}\right)\ \text{VS.}\ [S]_i\left(\frac{V_s}{V_o}\right)}$$

❑ A solution with successive additions



Successive standard addition should linear increase the analytical signal to 1.5 ~ 3 times to its origin values (B = 1.5A ~ 3A)

❑ Multiple solutions with same final volume

$$\frac{[X]_i}{[X]_f+[S]_f}=\frac{I_X}{I_{(X+S)}}$$

V_o and V are constant
V_S is variable

$$I_{(X+S)}=\frac{I_X}{[X]_i}\left([X]_f+[S]_f\right)$$

$$\underbrace{I_{(X+S)}}_{y}=\frac{I_X}{[X]_i}[X]_f+\frac{I_X}{[X]_i}\underbrace{[S]_f}_{x}$$

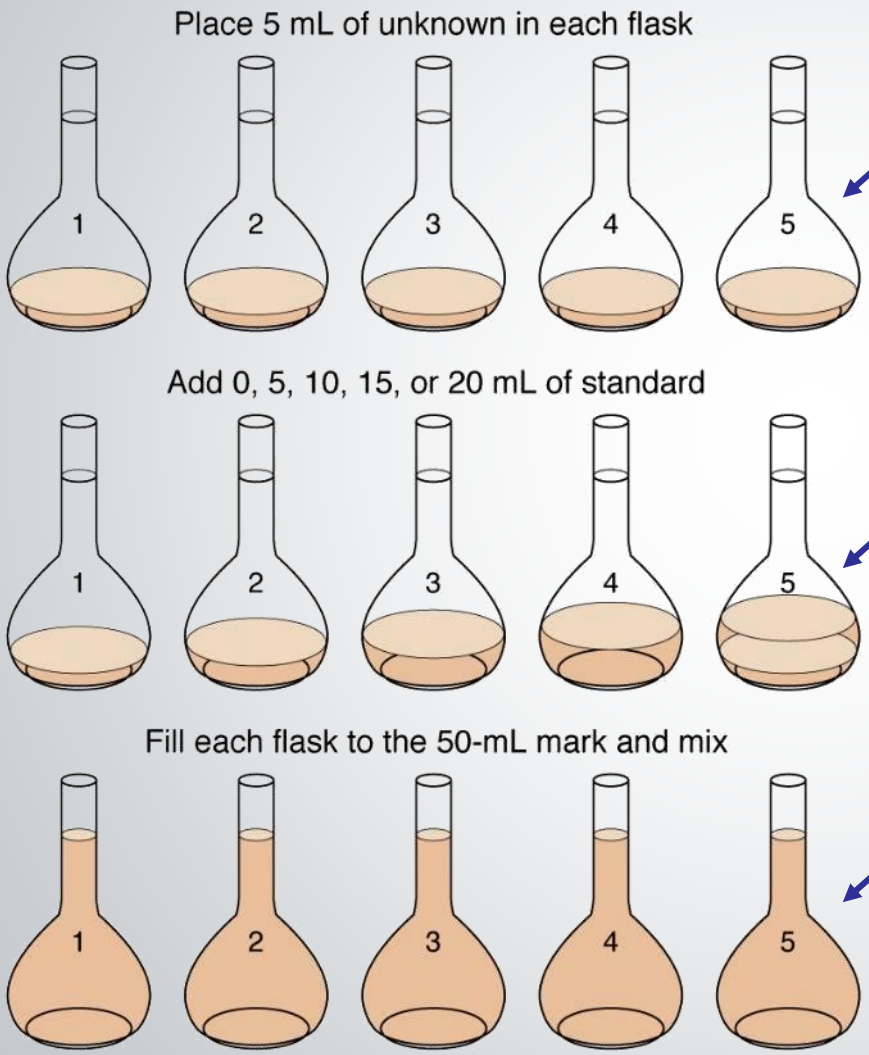
$$[S]_f=[S]_i\frac{V_S}{V}$$

$$V=V_o+V_s \quad [X]_f=[X]_i\frac{V_o}{V}$$

- Standard addition into several volumetric flask with constant V, and plot with

$$I_{(X+S)} \text{ VS. } [S]_f$$

❑ Multiple solutions with same final volume

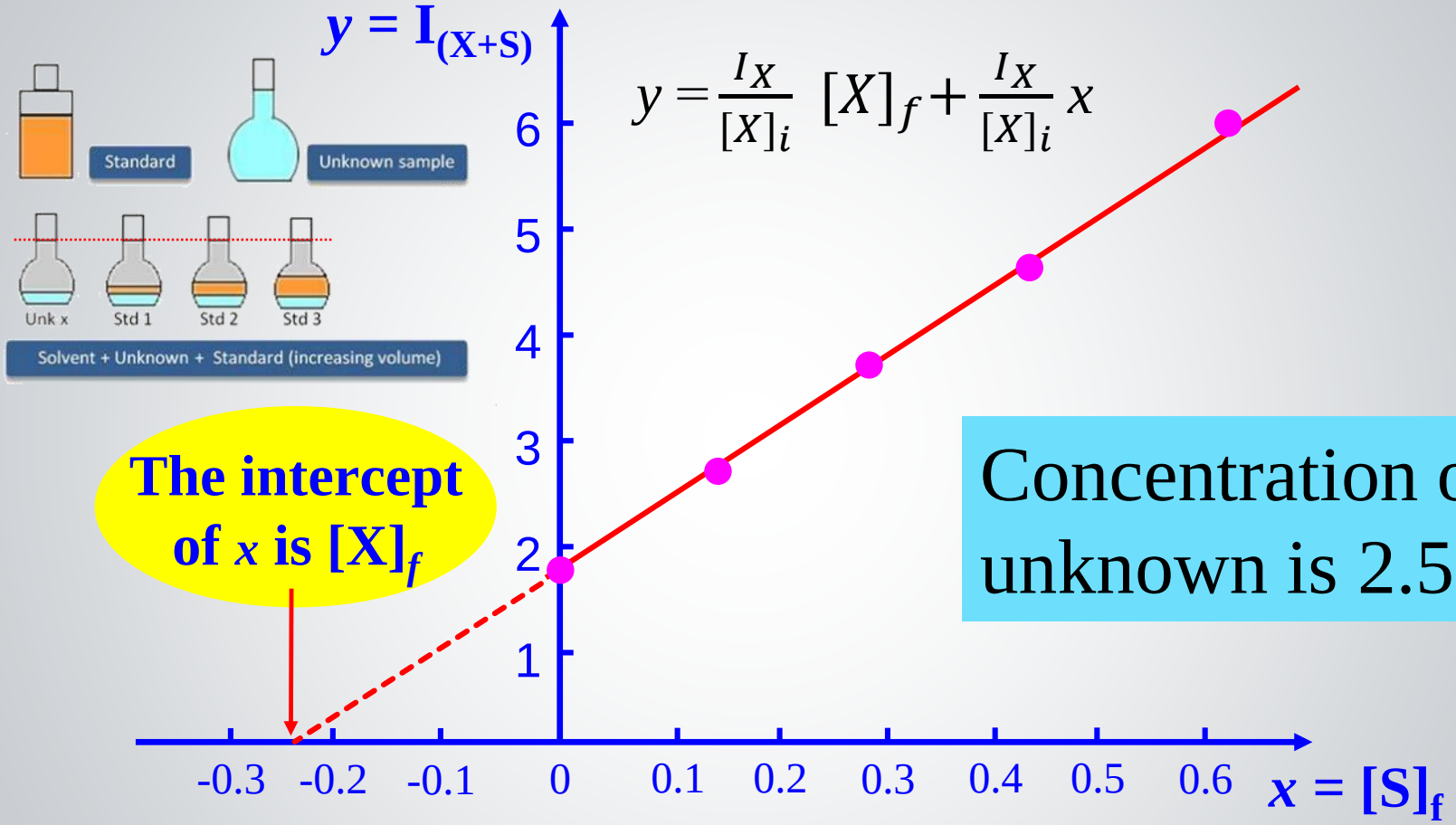


Add V_o=5 mL of unknown

Add standard of analyte [C] with increase V_s

Fill all to V=50 mL

❑ Multiple solutions with same final volume



Concentration of unknown is 2.5 M

Question: 5 mL of unknown in each flask are diluting to 50 mL. If X-intercept is 0.25 M, how much [C] for the unknown sample?

Comparisons with standard addition methods

	A solution with successive additions	Multiple solutions with constant V
Adding standard	Make a series of additions of standard with small V_s to the unknown sample	Place equal quantities of unknown into several volumetric flasks. Add an increase standard to each flask. Dilute them to the same final V.
Measuring	Measure signal after each addition. (increase signal by 1.5 to 3 times)	Measure signal for each of solution
***x* - *y* meaning**	Corrected signal VS. corrected $(\frac{V_s}{V_o})[S]_i$ of added standard	Signal VS. final $[S]_f$ of added standard after dilution
Find the unknown	*x*-intercept is **original $[X]_i$**	*x*-intercept is **$[X]_f$ in flask,** compute $[X]_i$ through **dilution factor**

1.3-2 Calibration methods

It is a method to measure the unknown quantity from the known quantities of analyte (standards)

① External standard

② Standard addition

③ Internal standard (learn by yourselves)

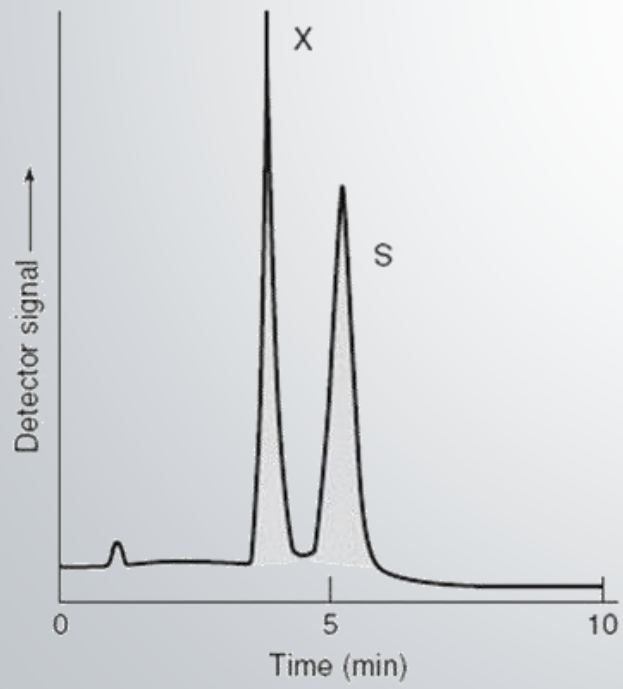
③ Internal Standard

It is a known amount of compound (Standard, S), –**different from** analyte (X)–add to the unknown.

- **Chemically similar between internal standard and analyte, don't react with each other;**
- **Use in uncontrolled sample lost occurs during sample preparation or analytical procedures;**
- **The instrument response varies slightly.**

One point of internal standard addition

- Preliminary experiment, 0.0837 M of X and 0.0666 M of S gave peak areas of A_X= 423 and A_S= 347
- 10.0 mL of 0.146 M of S were added to 10.0 mL of unknown, then diluted to 25.0 mL. The results were A_X= 553 and A_S= 582.



$$\frac{A_X}{[X]} = F\left(\frac{A_S}{[S]}\right)$$

A_X: signal from analyte
[X]: concentration of analye
A_S: signal from standard
[S]: concentration of standard
F : Correction factor

- Preliminary experiment, 0.0837 M of X and 0.0666 M of S gave peak areas of A_X= 423 and A_S= 347
- 10.0 mL of 0.146 M of S were added to 10.0 mL of unknown, then diluted to 25.0 mL. The results were A_X= 553 and A_S= 582

$$\frac{423}{0.0837\ \mathrm{M}} = F\left(\frac{347}{0.0666}\right) \Rightarrow F = 0.970_0 \qquad \frac{A_X}{[X]} = F\left(\frac{A_S}{[S]}\right)$$

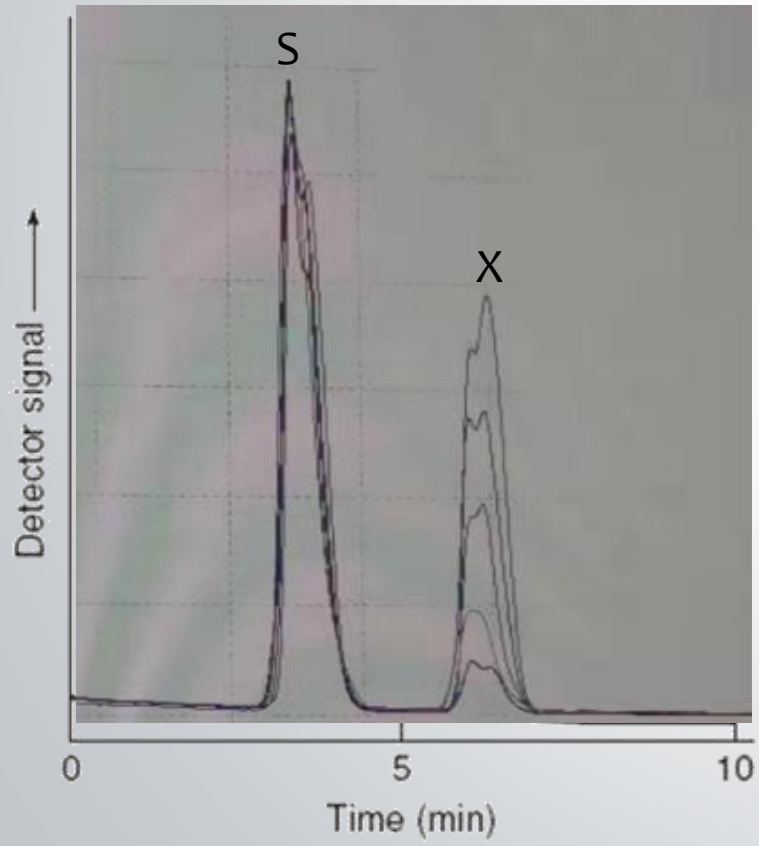
$$[S] = \frac{10.0\ mL}{25.0\ \mathrm{mL}} \times 0.146\ \mathrm{M} = 0.0584\ \mathrm{M}$$

$$\frac{553}{[X]} = 0.970_0\left(\frac{582}{0.0584}\right) \Rightarrow [X] = 0.0572_1\ \mathrm{M}$$

[X] of the unknown is $0.0572_1 \times \frac{25.0\ mL}{10.0\ \mathrm{mL}}\ \mathrm{M} = 0.184\ \mathrm{M}$

❑ Multiple solutions of internal standard addition

Linear regression affords a straight line that follows the equation:



R=mC+b

$$R = \frac{\text{area of analyte peak}}{\text{area of internal standard peak}}$$

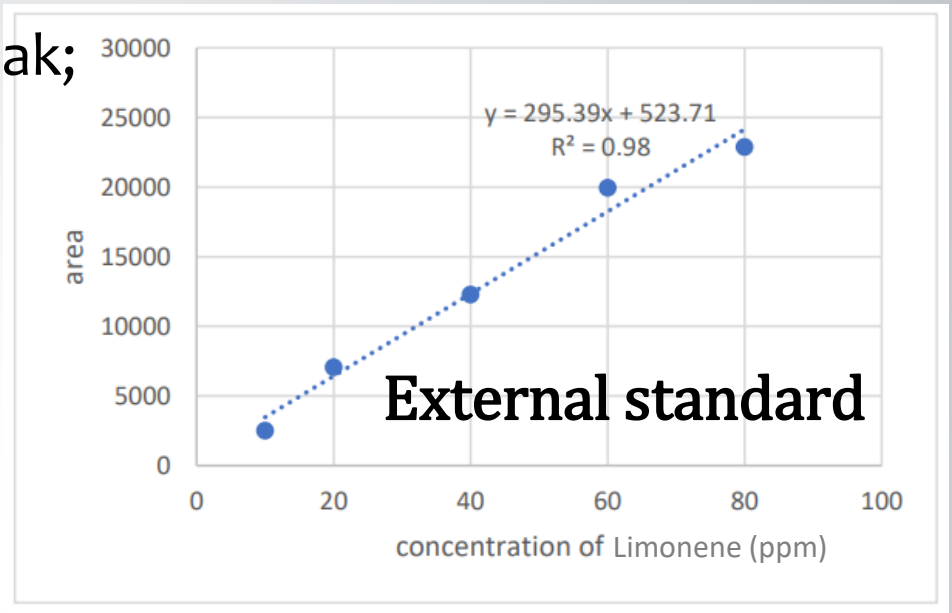
m = slope

C = analyte concentration

b = point of intercept with Y axis

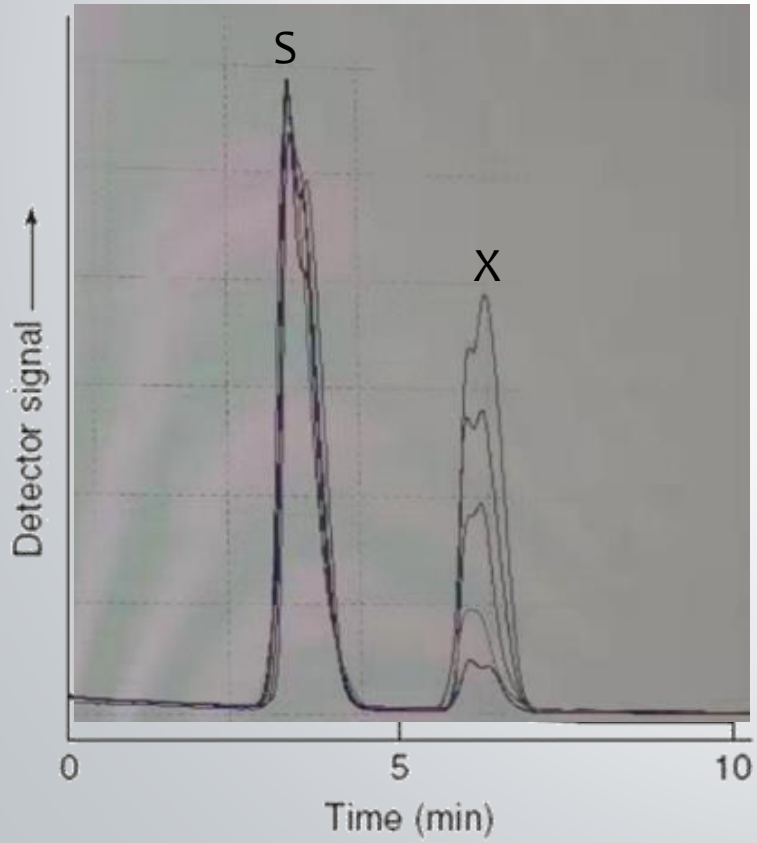
❑ Multiple solutions of internal standard addition

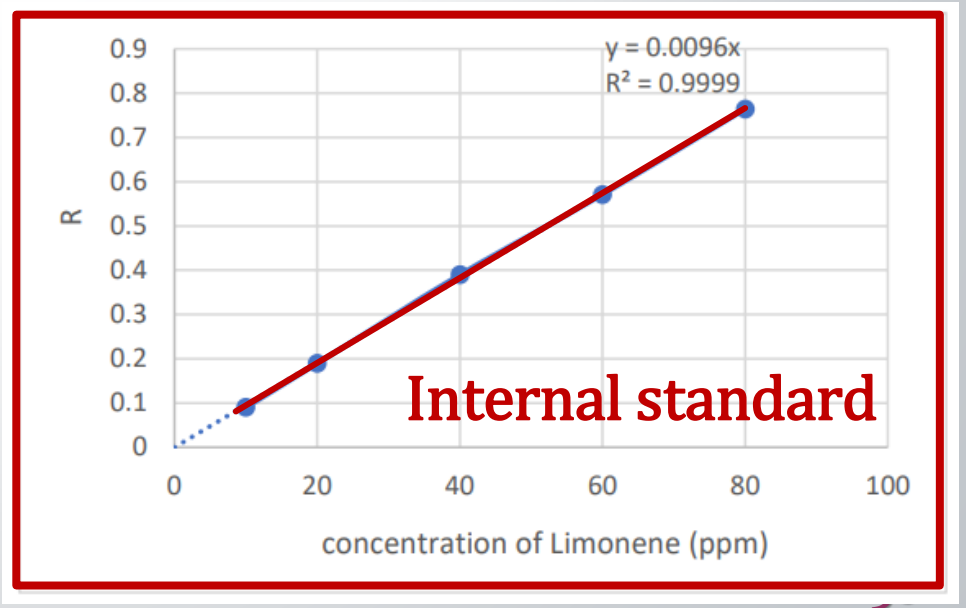
y axis represents Area of analyte peak;



y axis represents the ratio (R):

$$R = \frac{\text{Area of analyte peak}}{\text{Area of stadard peak}}$$





Comparisons with calibration methods

Approaches	Comparison
External standard	Analyte with known [C] used to prepare a **calibration curve** to deduce the unknown
Standard addition	Added standard **is the same substance** as analyte to address matrix effect
Internal Standard	Added standard is **different from analyte** when the quantity of analyte is not reproducible or sample losses occur